

Xinwen (Ellen) Zhang

Ph.D. Student at Temple University

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EDUCATION

Ph.D. in Computer and Information Science

Aug. 2022 – Present

Temple University, Philadelphia, USA

Advisor: Dr. Hongchang Gao

B.S. in Mathematics

Sep. 2017 – Jun. 2021

Tongji University, Shanghai, China

RESEARCH INTERESTS

Optimization for Large-Scale Machine Learning with a special focus on

Compositional Optimization for AUC Maximization, Optimization Beyond Minimization, and Optimization for Large Language Models.

HONORS AND AWARDS

- Outstanding Graduate Research Assistant Award, Temple University 2026
- ICML Financial Aid Award 2026
- ICML Gold Reviewer 2026
- Female Student Award in Data Science, IEEE International Conference on Data Mining 2025
- ICDM Student Travel Award 2025
- ICML Travel Award 2024
- WiML Travel Funding 2023
- NeurIPS Scholar Award 2023
- KDD Student Travel Award 2023
- Second Prize, Shanghai Mathematical Model Competition 2018
- Academic Scholarships, Three Consecutive Years at Tongji University 2017–2021

SELECTED PUBLICATIONS

[ICDM'26] **Xinwen Zhang**, Liang Zhan, Hongchang Gao. “Faster Federated Smoothed Minimax Optimization.” *In IEEE International Conference on Data Mining, 2026.*

[ICML'26] **Xinwen Zhang**, Richard Souvenir, Hongchang Gao. “LS²MC-GDA: A Smoothed Algorithm for Federated Stochastic Multi-level Compositional Minimax Optimization.” *In Proceedings of the 43rd International Conference on Machine Learning, 2026.*

[ICML'26] **Xinwen Zhang**, Yihan Zhang, Hongchang Gao, Heng Huang. “Distributed Stochastic K-Level Optimization Over Networks.” *In Proceedings of the 43rd International Conference on Machine Learning, 2026.*

[ICML'26] Yihan Zhang, **Xinwen Zhang**, My T. Thai, Jie Wu, Hongchang Gao. “Convergence Analysis of Decentralized Hessian-/Jacobian-Free Algorithm for Nonconvex Stochastic Bilevel Optimization.” *In Proceedings of the 43rd International Conference on Machine Learning, 2026.*

[ToN'26] Jiyao Liu, **Xinwen Zhang**, Xinliang Wei, Xuanzhang Liu, Yuzhou Chen, Hongchang Gao, Yu Wang. “Swapping and Purification Scheme Optimization for Entanglement Distribution in Quantum Networks.” *IEEE*

Transactions on Networking, 2026.

[NeurIPS'25] **Xinwen Zhang**, Hongchang Gao. "On the Convergence of Stochastic Smoothed Multi-Level Compositional Gradient Descent Ascent." *In Proceedings of the 39th Conference on Neural Information Processing Systems, 2025.*

[ICDM'25] **Xinwen Zhang**, Hongchang Gao. "Sharpness-Aware Optimization Through Variance Suppression on Deep AUC Maximization." *In IEEE International Conference on Data Mining, 2025.*

[IWQoS'25] Jiyao Liu, **Xinwen Zhang**, Xinliang Wei, Xuanzhang Liu, Yuzhou Chen, Hongchang Gao, Yu Wang. "Joint Swapping and Purification with Failures for Entanglement Distribution in Quantum Networks." *In IEEE/ACM 33rd International Symposium on Quality of Service, 2025.*

[ICML'24] **Xinwen Zhang**, Ali Payani, Myungjin Lee, Richard Souvenir, Hongchang Gao. "A Federated Stochastic Multi-Level Compositional Minimax Algorithm for Deep AUC Maximization." *In Proceedings of the 41st International Conference on Machine Learning, 2024.*

[NeurIPS'23] **Xinwen Zhang**, Yihan Zhang, Tianbao Yang, Richard Souvenir, Hongchang Gao. "Federated Compositional Deep AUC Maximization." *In Proceedings of the 37th Conference on Neural Information Processing Systems, 2023.*

Preprints

[arXiv'25] **Xinwen Zhang**, Yihan Zhang, Heng Liang, Hongchang Gao. "Nonconvex Decentralized Stochastic Bilevel Optimization under Heavy-Tailed Noises." *arXiv preprint arXiv:2509.15543.*

[arXiv'25] **Xinwen Zhang**, Hongchang Gao. "On Provable Benefits of Muon in Federated Learning." *arXiv preprint arXiv:2510.03866.*

[arXiv'25] **Xinwen Zhang**, Hongchang Gao. "Federated Stochastic Minimax Optimization under Heavy-Tailed Noises." *arXiv preprint arXiv:2511.04456.*

TUTORIALS & WORKSHOPS

[ICDM'25 Tutorial] Hongchang Gao, **Xinwen Zhang**. "Federated Stochastic Compositional and Bilevel Optimization." *In 2025 IEEE International Conference on Data Mining.*

[IJCAI'25 Tutorial] Hongchang Gao, **Xinwen Zhang**. "Federated Stochastic Compositional and Bilevel Optimization." *In Proceedings of the 34th International Joint Conference on Artificial Intelligence, 2025.*

[WiML'23 Workshop] **Xinwen Zhang**, Richard Souvenir, Hongchang Gao. "A Fast Federated Stochastic Compositional Minimax Optimization Algorithm with Variance Reduction." *In 18th Women in Machine Learning Workshop, 2023.*

[KDD'23 Tutorial] Hongchang Gao, **Xinwen Zhang**. "Distributed Optimization for Big Data Analytics: Beyond Minimization." *In Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2023.*

WORK EXPERIENCE

Teaching Assistant

Aug. 2022 – Dec. 2024

Department of Computer and Information Sciences, Temple University, USA

- CIS3715 Principles of Data Science Spring 2023, 2024
- CIS1051 Introduction to Problem Solving and Programming in Python Fall 2022, 2023, 2024,
Spring 2023, 2024

Data Analysis Intern

Mar. 2021 – Jun. 2021

Zhongyuan Consumer Finance, Shanghai, China

- Built Python/SQL pipelines and profiling models for credit monitoring and targeted marketing.
- Analyzed festival campaign data, showing cash incentives improved customer retention.

ACADEMIC SERVICES

Conference Reviewer

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|---|------|
| • AAAI | 2027 |
| • AAAI, ICML, NeurIPS, KDD, ICLR, AISTATS, CVPR | 2026 |
| • AAAI, ICML, NeurIPS, KDD, ICLR, AISTATS | 2025 |
| • AAAI, ICML, NeurIPS | 2024 |

Journal Reviewer

- ACM Transactions on Computing for Healthcare
- Journal of Combinatorial Optimization
- Journal of Optimization Theory and Applications
- Neural Networks
- Transactions on Machine Learning Research